



Multi-objective optimization of the cathode catalyst layer micro-composition of polymer electrolyte membrane fuel cells using a multi-scale, two-phase fuel cell model and data-driven surrogates

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ABSTRACT

Polymer electrolyte membrane fuel cells (PEMFCs) are considered a promising alternative to internal combustion engines in the automotive sector. Their commercialization is mainly hindered due to the cost and effectiveness of using platinum (Pt) in them. The cathode catalyst layer (CL) is considered a core component in PEMFCs, and its composition often considerably affects the cell performance (V_{cell}) also PEMFC fabrication and production (C_{stack}) costs. In this study, a data-driven multi-objective optimization analysis is conducted to effectively evaluate the effects of various cathode CL compositions on V_{cell} and C_{stack} . Four essential cathode CL parameters, i.e., platinum loading (L_{Pt}), weight ratio of ionomer to carbon ($wt_{\text{I/C}}$), weight ratio of Pt to carbon ($wt_{\text{Pt/C}}$), and porosity of cathode CL (ε_{CL}), are considered as the design variables. The simulation results of a three-dimensional, multi-scale, two-phase comprehensive PEMFC model are used to train and test two famous surrogates: multi-layer perceptron (MLP) and response surface analysis (RSA). Their accuracies are verified using root mean square error and adjusted R^2 . MLP which outperforms RSA in terms of prediction capability is then linked to a multi-objective non-dominated sorting genetic algorithm II. Compared to a typical PEMFC stack, the results of the optimal study show that the single-cell voltage, V_{cell} is improved by 28 mV for the same stack price and the stack cost evaluated through the U.S department of energy cost model is reduced by \$5.86/kW for the same stack performance.

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1. Introduction

Fuel cells (FCs) are devices that electrochemically combine fuel, such as hydrogen and oxygen, to generate electricity along with water and heat. Electricity is continuously generated if the fuel source is constantly supplied. FCs have no moving parts and do not burn fuels, which makes their operation extremely quiet and pollution free. Based on the materials used for construction, FCs are mainly categorized into six groups: polymer electrolyte membrane FC (PEMFC), alkaline FC, direct methanol FC, phosphoric acid FC, solid oxide FC, and molten carbonate FC [1]. Among the FCs, PEMFCs have been considered as one of the most promising alternatives to the internal combustion (IC) engines of automobiles. Compared to IC engines, PEMFCs have zero carbon emissions, oper-

ate at low operating temperatures, are easy to start, and have efficiencies of around 60% [2].

A typical PEMFC comprises of multiple FCs that are connected in series to form a stack. The core component of a PEMFC is the membrane electrode assembly (MEA), which includes two catalyst layers (CLs), a polymer electrolyte membrane, and gas diffusion layers (GDLs). MEA is one of the primary locations where electrochemical reactions, such as hydrogen oxidation reaction (HOR) in the anode and oxygen reduction reaction (ORR) in the cathode, occur. Moreover, the multiphase transport of fuel and oxidant, transfer of protons from the anode to the cathode through the membrane, and transport of liquid water generated in the cathode takes place in the MEA [3]. Consequently, numerous researchers are focused on developing and upgrading the MEA to improve the PEMFC performance and durability and to reduce costs, with a particular emphasis on low catalyst loading and good endurance under real-world conditions [4–6].

Although PEMFCs have grown in popularity over the last decade, durability and cost have impeded their commercialization.

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Nomenclature

a	Ratio of active surface area per unit electrode volume, m^2/m^3 or water activity	$\bar{y}$	Mean value of the observed data
A	Area, m^2	z	Transport resistance coefficient
B	Weight coefficient vector	<i>Greek symbols</i>	
C	Molar concentration of species, mol/m^3 or acceleration factor or fabrication and production cost, $\$/\text{kW}$	α	Transfer coefficient
d	Number of input variables	β	Weight coefficient
D	Species diffusivity, m^2/s	γ	Reaction order
E	Activation energy, kJ/mol or error function	δ	Thickness, m
EW	Equivalent weight of a dry membrane, kg/mol	ε	Volume fraction or error
f	Actual response function or objective function	η	Surface overpotential, V or learning rate
F	Faraday's constant, $96,487\text{C}/\text{mol}$ or optimization problem or pareto front	θ	Contact angle of the gas diffusion layer
G	Global best	λ	Water content
g	Constraint condition	$\hat{\mu}$	Estimated mean value from training data set distribution
h	Output of the hidden layer upon activation or constraint condition	κ	Proton conductivity, S/m
i_0	Exchange current density, A/cm^2	ϕ	Phase potential, V
i_d	Density estimation parameter	ρ	Density, kg/m^3
I	Operating current density, A/cm^2	σ	Electronic conductivity, S/m
j	Transfer current density, A/cm^2	τ	Viscous shear stress, N/m^2
k	Thermal conductivity, $\text{W}/\text{m}\cdot\text{K}$, or Relative permeability	ξ	Stoichiometry flow ratio
K	Hydraulic permeability, m^2	Ω	Oxygen transport resistance
L	Least square function or amount of loading, mg/cm^2	<i>Superscripts</i>	
n	Number of electrons transferred in the electrode reaction	eff	Effective
net	Product of weights and the training data	g	Gas
N	Number of training points	l	Liquid
M_w	Molecular weight, kg/mol	mem	Membrane
P	Pressure, Pa , or Particle personal best or population of parents	ref	Reference value
p	Number of weight coefficients in each second-order equation or number of weights in the hidden layer	<i>Subscripts</i>	
q	Mating rate	a	Anode
Q	Number of the sampling points or population of offsprings	C	Carbon
R	Total population of parents and offsprings	CL	Catalyst layer
s	Liquid saturation	c	Cathode
S	Source term in the transport equation or Population or dominated individual set	cCL	Cathode catalyst layer
t	Iteration time step	e	Electrolyte
T	Temperature, K	ch	Gas channel
$\vec{u}$	Fluid velocity and superficial velocity in a porous medium, m/s	gdl	Gas diffusion layer
V	Particle velocity or voltage, V	I/C	Ionomer to carbon weight ratio
W	Matrix of weights	i	Species
wt	Weight percentage	in	Channel inlet
X	Vector of the design variables or position vector	int	Interface
x	Input variable	mem	Membrane
y	Observed response	Pt/C	Weight ratio of Platinum to carbon
$\hat{y}$	Predicted response	Pt	Platinum
		s	Solid, surface
		T	Temperature
		u	Momentum equation
		w	Water
		0	Initial conditions or standard conditions, i.e., 298.15 K and 101.3 kPa (1 atm)

CLs are one of the most important components that determine an FC's durability and cost. CLs comprise various materials, including catalysts (mostly Pt), supporters (mostly carbon), and an ionomer for electrochemical reactions and resulting transport [7]. Since the electrochemical processes occur in the CL region, the characteristics of the CL region, such as chemical composition, porosity, and most crucially, catalyst loading, significantly impact the PEMFC performance. The ionomer serves as an electrolyte for proton conduction and a binder for CL. The ionomer loading within CL ranges between 20 and 40 wt%, where 30% loading of the ionomer has been best reported [8]. The mass ratio of ionomer to carbon sup-

port, or the $wt_{I/C}$ ratio, is commonly used to describe the Nafion concentration in CL. Note that when the $wt_{I/C}$ ratio is below the optimum, the low ionomer concentration causes a drop in the proton conduction and a loss in the cell performance. When the $wt_{I/C}$ ratio is above the optimum, the excess ionomer concentration restricts the space volume, thus reducing the efficiency of reactant supply and product removal as well as the cell performance [9]. The two-phase transport inside CL is determined by porosity, which is a significant characteristic. Pore formers, such as carbonates and oxalates, are widely employed to increase porosity by redistributing Pt/C particles. Ammonium carbonates and oxalates

can improve porosity from 35% to over 45%, while lithium carbonate can increase porosity up to 65%. However, increased porosity may negatively impact the FC efficiency due to the lower electrical conductivity induced by a poor catalyst–particle interaction [10,11].

Generally, the high cost of the PEMFC catalyst (usually Pt), which is extensively used in CLs, accounts for about 45% of the FC cost [12]. Pt is widely used as a catalyst due to its chemical stability and high exchange current density. As described above, CL comprises Pt particles dispersed within a complex porous matrix of carbon and electrolyte. According to the literature, the overall performance of the cell does not linearly increase with the amount of Pt dispersed with this complex matrix [13]. Rather, the cell performance is highly influenced by how effectively the Pt particles are dispersed with the complex porous matrix as it determines how many of them are available for the electrochemical reaction [14]. Since the operation of PEMFCs involves a triple phase boundary between the electrolyte, Pt, and carbon, the determination of the optimum composition for the electrochemical reactions becomes a major task. Furthermore, as the loading of the expensive electrocatalyst is decreased in state-of-the-art FCs, their durability decreases, which makes it difficult to meet the durability and cost targets set by the U.S department of energy (DOE) [15]. Since Pt catalyst is expensive and readily poisoned, alternate catalysts need to be found. Thus, finding an appropriate catalyst that fits certain parameters, such as high performance, low cost, and high durability, is critical in the development and commercialization of PEMFCs. Developing new electrodes with lower loads or no Pt is still a research hot topic, and more computational and experimental research is needed to achieve this goal [16,17]. There is a wide variety of catalysts that are currently being explored, for example, the transition metal–carbon electrode (M–N–C), transition metal oxides, nitrides, and carbides etc. [18]. Among these, the Fe–N–C has shown promising results in terms of catalytic activity [19–21]. Although replacing Pt with non-Pt particles, reduces the overall cost of PEMFCs, non-Pt catalysts are less recognized and offer lower performance [22,23].

The aforementioned studies have been conducted through experimental procedures, which are time consuming and expensive. Additionally, these experimental procedures are based on trial-and-error procedures and can explore a limited design space in a limited amount of time. Computational fluid dynamics (CFD) simulations aid in identifying limiting factors and decreasing the time and costs of the testing [24,25]. Moreover, they provide an insight into the phenomena occurring in PEMFCs at the microscopic and macroscopic levels. Kamarajugadda et al. [26] studied the effects of Nafion loading, Pt loading, agglomerate size, CL thickness (δ_{CL}), and Pt/C ratio on cell performance considering an agglomerate model. Their results, which were validated with experimental data, revealed that the cell performance trends are different at intermediate and high operating current densities. Hence, they suggest that there is an optimal value for Nafion and Pt content for optimizing the cell performance and cost of PEMFCs. Lee et al. [27] studied the effect of porosity on CLs and revealed that the cathode CL porosity considerably affects the cell performance compared to the anode CL porosity. This variation in performance can be attributed to the effective water removal rates at high current densities.

The literature survey provides an insight into the impact of CL design parameters, such as porosity, L_{Pt} , $wt_{I/C}$, and $wt_{Pt/C}$, on the cell performance. Although the application of computational models alleviates the costs incurred through experiments in the design loop, most numerical models are limited to the CL region and do not consider the necessary interaction with neighboring PEMFC components, such as membrane, micro-porous layers (MPL), and GDLs. Owing to physical complexities and computational

challenges, the development of a comprehensive, reliable, real-scale, PEMFC model that accounts for the microstructure of CLs has been rarely reported in the literature [28,29]. Alternatively, simple, and cheap approaches using surrogates or meta-models that can approximate the PEMFC behaviors have been developed in recent years [30–32]. Some of the well-known surrogates include artificial neural networks, support vector machine (SVM), Kriging (KRG), RSA, and random forest. Wang et al. [33] employed a SVM surrogate model coupled with a genetic algorithm (GA) to search for optimal CL compositions. Moreover, Zhang et al. [34] used SVM as a surrogate model to improve the porous media flow field shape of a PEMFC. Their optimization results, which were confirmed by a three-dimension (3D) physical model, demonstrated the capability of the data-driven surrogate model for optimizing the PEMFC components. Xing et al. [35] used simulation data from a two-dimensional CFD model to yield a KRG surrogate model, which was then integrated with an optimization technique to identify the global optimum design of CLs. Furthermore, Cheng et al. [36] proposed an optimization methodology to identify the ideal operating parameters of a PEMFC using a radial basis function (RBF) neural network surrogate model. They analyzed the effects of operational parameters on the PEMFC performance using the design of experiments approach and Taguchi orthogonal array method. Additionally, they verified the accuracy of the RBF neural network model using a cross-validation technique based on significant components and analysis of variance.

The above literature survey shows that the optimum composition and structure of the CLs have been traditionally determined using costly and time-consuming CL experiments. Computer simulations make it easier to experiment with different ideas as well as save cost and time by reducing the need for pricey prototypes. Although cheaper data-driven surrogates and metamodels can yield numerical results faster and are more cost-effective compared to CFD-based numerical models, constructing reliable sample data is a prerequisite for ensuring the accuracy of model predictions. In addition, as described by Simon et al. [37], the PEMFC stack accounts for approximately 45% of the total system cost, where one of the most significant contributors to a high price in stacks is the amount of catalyst (Pt) used in them. Therefore, in CL design and optimization, the cost aspect of a PEMFC stack should be considered along with the performance to promote the commercialization of PEMFC technologies. However, to date most CL design and optimization studies have considered only single-objective optimization, mainly maximizing CL performance. Motivated by the present shortcoming, a multi-objective optimization study considering key nano- or micro-scale CL design parameters is performed for the first time where increasing cell performance and decreasing the stack cost simultaneously becomes the primary objective.

In this study, we attempt to optimize key micro-compositions in the cathode CL using a CFD-based optimization approach. First, under suitable constrain conditions and varying micro-compositions of the CL, sample points are drawn using the Latin hypercube sampling (LHS) technique. At the generated sample points, a database of cell performance via CFD simulations is constructed considering the multi-scale, two-phase, 3D comprehensive PEMFC model developed in our previous study [29,38–40]. Then, considering the database, two famous surrogates, i.e., MLP and RSA, are constructed, trained, and tested using MATLAB R2021b. Moreover, to determine the optimal cathode CL composition, the surrogates are further coupled with the particle swarm optimization (PSO) algorithm. The accuracy of the optimum microscopic CL design values is further evaluated via comprehensive PEMFC model simulations to probe the validity of such FC optimization methodology. Additionally, to effectively evaluate the cost of a 67.5 kW PEMFC stack for a typical FC road vehicle, the

DOE cost model is employed [37]. Furthermore, the trained surrogates and DOE cost model are linked with a sophisticated multi-objective NSGA II to determine the optimal 67.5 kW PEMFC stack with the least cost.

2. Multi-scale PEMFC model

A 3D, multi-scale, two-phase PEMFC model based on the multi-phase mixture (M^2) approach proposed by Wang and Cheng [41] is considered in this study. Modeling of the PEMFCs is conducted considering various components, such as the membrane, CLs, GDLs, and bipolar plates (BPs). The 3D PEMFC model has been previously validated against experimental polarization curves measured under various cell designs and operating conditions [38]. The model assumptions and governing equations considered herein are similar to those described in our previous studies [38,40,42]. Therefore, only a summary of the same has been presented in Sections 2.1 and 2.2, respectively. Additionally, the boundary conditions and numerical implementations of the PEMFC model featured using commercial CFD software ANSYS Fluent (ANSYS Inc., USA) are summarized in Section 2.3.

2.1. Model assumptions

The following specific assumptions are considered in this study:

1. The operating pressures considered are low, and hence, ideal gas mixtures are considered in the gas phase.
2. The flow velocity is low and laminar.
3. The effects of gravity are negligible.
4. In the porous regions, the effect of immobile liquid saturation is negligible.
5. Pt particles in CLs are assumed to have the spherical shape of the same size.

2.2. Governing equations and source terms

The 3D PEMFC model considered in this study is governed by five conservation principles: mass, species, charge, momentum, and thermal energy. The conservation equations are coupled with source terms related to the electrochemical reactions in the anode (HOR) and cathode (ORR). Table 1 and Table 2 list the governing equations and source terms.

Additionally, Table 3 summarizes a list of equations that are related to the kinetic and physicochemical properties of PEMFC components. The mixture properties related to the M^2 model are listed in Table 4. Furthermore, in Table 5, the species and transport

properties that are correlated to the water content λ , which is a function of the water activity a [43], are summarized.

2.3. Microscale CL model

Table 6 lists the parameters of a CL particle in a microscale CL domain, including the volume fractions of Pt (ε_{Pt}), carbon (ε_C), and ionomer (ε_e) as well as δ_{CL} . The aforementioned parameters are based on the ionomer to carbon weight ratio (I/C), weight percent of Pt/C, and Pt loading function. Since the ionomer layer is hydrophilic, a layer of water covers it, and the thicknesses of the ionomer and water films are determined by Eqs. (38) and (39), respectively. Additionally, using Eqs. (41)–(43), the oxygen flux across the water and ionomer films, the volumetric surface area of the ionomer (a_c), j_c for ORR, the active volumetric surface area of Pt (a_{Pt}), and the electrochemical active surface area (ECSA), are determined. The interfacial resistance in the water and ionomer films as well as the effective thicknesses of the water and ionomer films (δ_w^{eff} and δ_e^{eff} , respectively) are used to calculate the total oxygen transport resistance, represented by the variable Ω_T . For further insight into the microscale modeling, please refer to our previous studies [29,39].

2.4. Boundary conditions and numerical implementation

Fig. 1 shows the schematic of the PEMFC stack and unit cell structures along with detailed dimensions and boundary conditions of the single-cell PEMFC analyzed herein. The anode and cathode inlet velocities that are calculated using the stoichiometric ratios (ξ_a and ξ_c , respectively) are described. In terms of the mass flow, no-slip and impermeability criteria are applied to all the exterior surfaces except the inlet and outlet areas of the anode and cathode gas channels. For temperature calculations, an isothermal boundary condition is applied to the side walls. The PEMFC can be operated under either galvanostatic or potentiostatic mode. Hence, either a constant voltage or current density is applied at the outer sidewall of the cathode BP and the electric potential ϕ_s is assumed to be zero on the outer surface of the anode BP, as shown in Fig. 1 (b). A grid independence study is conducted to determine the optimal number of meshes. The results are shown in Fig. S1 where the optimum number of grid points is estimated to be around 280,000 based on the analysis accuracy and time necessary. Using user-defined functions, the PEMFC model is numerically incorporated into the commercially accessible CFD program ANSYS Fluent ver. 19 (ANSYS, Inc., US). The convergence criterion for the equation residuals is set at 10^{-8} .

Table 1
PEMFC model: Governing equations.

Governing equations		
Mass	$\nabla \cdot (\rho \vec{u}) = 0$	(1)
Momentum	$\left(\frac{1}{\rho}\right) \nabla \cdot (\rho \vec{u} \vec{u}) = -\nabla P + \nabla \cdot \tau + S_u$	(2)
Species	Flow channels and porous media: $\nabla \cdot (\gamma_i \rho m_i \vec{u}) = \nabla \cdot [\rho^g D_i^{g,eff} \nabla (m_i^g)] + \nabla \cdot [(m_i^g - m_i^l) \vec{j}^i] + S_i$ Water transport in the membrane: $\nabla \cdot \left(\left(\frac{\rho_w^{mem}}{EW} \right) D_w^{mem} \nabla \lambda \right) M_w - \nabla \cdot (n_d \left(\frac{F}{e} \right) M_w) + \nabla \cdot \left(\left(\frac{k_w^{mem}}{v} \right) \nabla P^l \right) = 0$	(3) (4)
Charge	Proton transport: $\nabla \cdot (k^{eff} \nabla \phi_e) + S_\phi = 0$ Electron transport: $\nabla \cdot (s^{eff} \nabla \phi_s) - S_\phi = 0$	(5) (6)
Energy	$\nabla \cdot (\rho \vec{u} C_p^g T) = \nabla \cdot (k^{eff} \nabla T) + S_T$	(7)

Table 2
PEMFC model: Source and sink terms.

Description	Expressions
Momentum	Porous media $S_{it} = -\frac{\mu}{k} \vec{u}$
Species	H ₂ in anode CL $S_{H_2,a} = \left[-\frac{j}{2F}\right] M_{H_2}$ O ₂ in cathode CL $S_{O_2,c} = \left[\frac{j}{4F}\right] M_{O_2}$ Water in anode CL $S_{w,a} = \left[-\nabla \cdot \left(\frac{n_w}{F} \vec{T}\right) + \frac{j}{4F}\right] M_w$ Water in cathode CL $S_{w,c} = \left[-\nabla \cdot \left(\frac{n_w}{F} \vec{T}\right) - \frac{j}{2F}\right] M_w$
Energy	In anode CL $S_{T,a} = j \cdot \eta + \frac{j^2}{k_{eff}}$ In cathode CL $S_{T,c} = j \left(\eta + T \frac{d\eta}{dT} \right) + \frac{j^2}{k_{eff}}$ In membrane $S_T = \frac{j^2}{k_{eff}}$
Charge	In CLs: $S_\phi = j$
Electrochemical reactions $\sum_k s_i M_i^z = ne^-$, where $\begin{cases} M_i = \text{chemical formula of species} \\ s_i = \text{stoichiometric coefficient} \\ n = \text{number of electrons transferred} \end{cases}$	
HOR on the anode side: $H_2 - 2H^+ = 2e^-$	
ORR on the cathode side: $2H_2O - O_2 - 4H^+ = 4e^-$	
Transfer current density, $[A/m^2]$	HOR in anode CL: $j = (1-s) a_{0,a}^{ref} \left(\frac{C_{H_2}}{C_{H_2,ref}} \right)^{\frac{1}{2}} \exp \left[-\frac{E_a}{R} \left(\frac{1}{T} - \frac{1}{353.15} \right) \right] \left(\exp \left(\frac{\alpha_a F}{RT} \eta \right) - \exp \left(-\frac{\alpha_c F}{RT} \eta \right) \right)$ ORR in cathode CL: $j = -\frac{3L_m}{r_{Pt} \cdot \rho_{Pt} \cdot \alpha_{CL}} i_{0,c}^{ref} \left(\frac{C_{O_2}}{C_{O_2,ref}} \right)^{3/4} \exp \left(-\frac{E_c}{R} \left(\frac{1}{T} - \frac{1}{353.15} \right) \right) \exp \left(-\frac{\alpha_c F}{R_u T} F \eta \right)$
Overpotential	$\eta = \phi_s - \phi_e - U$ where $U = U_0 - \frac{RT}{nF} \ln \frac{C_{O_2}}{C_{O_2,ref}}$

Table 3
Kinetic, transport, and physiochemical properties.

Description	Value/Expression	Ref.
Exchange current density of HOR $\times$ ECSA per unit CL volume, $a_{0,a}^{ref}$	$1.2 \times 10^{10} A/m^3$	[44]
Exchange current density for ORR, $i_{0,c}^{ref}$	$2.0 \times 10^{-4} A/cm^2$ -Pt	[44]
Activation energy of anode, E_a	10.0 kJ/mol	[44]
Activation energy of cathode, E_c	70.0 kJ/mol	[44]
Transfer coefficient of HOR, $\alpha_a = \alpha_c$	1	[45]
Transfer coefficient of ORR, α_c	1	[45]
Reference H ₂ /O ₂ molar concentration, C^{ref}	40.88 mol/m ³	[44]
Permeability of GDL/CL, K_{GDL}/K_{CL}	$1.0 \times 10^{-12}/1.0 \times 10^{-13} m^2$	[46]
Equivalent weight of electrolyte in the membrane, EW	1.1 kg/mol	[47]
Faraday's constant, F	96,485C/mol	
Universal gas constant, R_u	8.314 J/(mol · K)	
H ₂ diffusivity in the anode gas channel, $D_{0,H_2,a}^g$	$1.1028 \times 10^{-4} m^2/s$	[47]
H ₂ O diffusivity in the anode gas channel, $D_{0,H_2O,a}^g$	$1.1028 \times 10^{-4} m^2/s$	[47]
O ₂ diffusivity in the cathode gas channel, $D_{0,O_2,c}^g$	$3.2348 \times 10^{-4} m^2/s$	[47]
H ₂ O diffusivity in the cathode gas channel, $D_{0,H_2O,c}^g$	$7.35 \times 10^{-5} m^2/s$	[47]
Binary gas diffusivity (D_i^g)	For nonporous regions $D_i^g = D_o^g \left(\frac{T}{T_o} \right)^{3/2} \left(\frac{p_o}{p} \right)$	(11)
Effective diffusivity ($D_i^{g,eff}$)	For porous regions $D_i^{g,eff} = \frac{\epsilon}{\tau_k} D_i^g$	(12)

3. Meta models and optimization

Surrogate models, which are often called black-box models, response surfaces, metamodels, and emulators are simplified analytical models that can mimic the input/output behaviors of complex systems or higher-order models. Surrogate models are often employed in the field of engineering design optimization where the use of experimental and computer simulations for optimization is time consuming and expensive. Data-driven surrogate models are developed through computationally expensive computer simulations at a set of carefully selected sample points. These mod-

els can approximate the behavior of the underlying sophisticated simulations and are computationally less expensive to analyze. Thus, surrogate models can be thought of as a simplified representation of a complicated system in a specified domain with relaxed precision. During the development of these models, the tradeoff between accuracy and processing time is essential.

Construction of data-driven surrogates aided by computer simulations for optimization comprises the following two important steps: selection of sample points to train and test the surrogates from a specified design space and construction and evaluation of the accuracy of the surrogates at the respective training and test

Table 4
Expressions used in the two-phase mixture model.

Description	Expression
Mixture density (ρ)	$\rho = \rho^l s + \rho^g (1 - s)$ (13)
Gas mixture density (ρ^g)	$\rho^g = \left(\frac{P}{R_g T} \right) \frac{1}{\sum_i \frac{m_i}{M_i}}$ (14)
Mixture velocity ($\vec{u}$)	$\vec{u} = (\rho^l \vec{u}^l + \rho^g \vec{u}^g) / \rho$ (15)
Mixture mass fraction	$m_i = \frac{\rho^l s m_i^l + \rho^g (1-s) m_i^g}{\rho}$ (16)
Relative permeability	$k_r^l = s^3 k_r^g = (1 - s)^3$ (17)
Kinematic viscosity of the two-phase mixture	$\nu = \left(\frac{k_r^l}{\nu^l} + \frac{k_r^g}{\nu^g} \right)^{-1}$ (18)
Kinematic viscosity of the gas mixture	$\nu^g = \frac{\mu^g}{\rho^g} = \frac{1}{\rho^g} \sum_{i=1}^n \frac{x_i \mu_i}{\sum_{j=1}^n x_j \phi_{ij}}$ where $\phi_{ij} = \frac{1}{\sqrt{8}} \left(1 + \frac{M_i}{M_j} \right)^{-1/2} \left[1 + \left(\frac{\mu_i}{\mu_j} \right)^{1/2} \left(\frac{M_j}{M_i} \right)^{1/4} \right]^2$ (20)
	and (21)
	$\mu_i [N.s.m^{-2}] = \begin{cases} \mu_{H_2} = 0.21 \times 10^{-6} T^{0.66} \\ \mu_w = 0.00584 \times 10^{-6} T^{1.29} \\ \mu_{N_2} = 0.237 \times 10^{-6} T^{0.76} \\ \mu_{O_2} = 0.246 \times 10^{-6} T^{0.78} \end{cases}, T \text{ in kelvin}$ (22)
Relative mobility	$\lambda^l = \frac{k_r^l}{\nu^l} \nu^g \lambda^g = 1 - \lambda^l$ (23)
Diffusive mass flux	$\vec{j}^l = \rho^l \vec{u}^l - \lambda^l \rho \vec{u} = \frac{k_r^l}{\nu^l} \lambda^l \lambda^g \nabla P_c$ (24)
Capillary pressure P_c	$P_c = P^g - P^l = \sigma \cos \theta \left(\frac{r}{R} \right)^{1/2} J(s)$ (25)
Leverett function $J(s)$	$J = \begin{cases} 1.417(1-s) - 2.120(1-s)^2 + 1.263(1-s)^3 & \text{if } \theta_c < 90^\circ \\ 1.417s - 2.120s^2 + 1.263s^3 & \text{if } \theta_c > 90^\circ \end{cases}$ (26)
	(27)

Table 5
Transport properties in the electrolyte.

Description	Expression
Membrane water content (λ)	$\lambda = \begin{cases} \lambda_g = 0.043 + 17.81a - 39.85a^2 + 36.0a^3 & \text{for } 0 < a \leq 1 \\ \lambda_l = 22 & \end{cases}$ (28)
	Water activity, $a = \frac{C_w R_g T}{P_{sat}}$ [43] (29)
Electro-osmotic drag (EOD) coefficient of water (n_d)	$n_d = \frac{2.5\lambda}{22}$ (30)
Proton conductivity (κ)	$\kappa = (0.5139\lambda - 0.326) \exp \left[1268 \left(\frac{1}{303} - \frac{1}{T} \right) \right]$ (31)
Water diffusion coefficient (D_w^{mem})	$D_w^{mem} = \begin{cases} 2.692661843 \cdot 10^{-10} \text{ for } \lambda \leq 2 \\ \{0.87(3-\lambda) + 2.95(\lambda-2)\} \cdot 10^{-10} \cdot e^{(7.9728-2416/T)} \text{ for } 2 < \lambda \leq 3 \\ \{2.95(4-\lambda) + 1.642454(\lambda-3)\} \cdot 10^{-10} \cdot e^{(7.9728-2416/T)} \text{ for } 3 < \lambda \leq 4 \\ \{2.563 - 0.33\lambda + 0.0264\lambda^2 - 0.000671\lambda^3\} \cdot 10^{-10} \cdot e^{(7.9728-2416/T)} \text{ for } 4 < \lambda \leq \lambda_{a=1}^g \end{cases}$ (32)
Interfacial resistance of the water film ($\Omega_{w,int}$)	$\Omega_{w,int} = z_w \frac{\delta_w}{D_{O_2,w}}$ (33)

points. Once the constructed surrogate has achieved the desired test and training accuracy, it is further linked to an optimization algorithm to obtain the global best solutions.

As mentioned earlier, constructing the surrogate models necessitates the generation of sample points in a selected design space; thus, a well-known sampling technique known as the LHS is employed. LHS was developed by McKay et al. [48] and is based on minimizing the maximum distance between sample points. Sample point generation is performed using a stratified technique where the range of sample points Q is explicitly split into Q intervals with identical and independent distribution. Fig. S2 presents the sample points generated through the LHS technique. Totally, 85 sample points were generated: 70 points were used for training and 15 points were used for testing the surrogate models. Considering the 70 training points, MLP and RSA are constructed and then tested against the 15 test points to verify the prediction accuracy of the models. The model's accuracy is determined by evaluating the root mean square error (RMSE) and adjusted R^2 at the respective training and test points. Both the error indicators are capable of conveying the goodness of fit differently, and their uniqueness will be further discussed in Section 3.2.

Once the models' prediction accuracy is verified, they are further linked to a stochastic single-objective optimization algorithm,

namely PSO, to determine the maximum global best value (max cell voltage (V_{cell})) and the optimal design variables at which PEMFCs afford the best operating performance. Furthermore, considering the optimal design points predicted by the surrogate models via PSO are validated by comparing them with the CFD simulation results.

Generally, an engineering optimization problem comprises two or more optimizing goals or objectives that are conflicting and need to be achieved; such problems are called multi-objective optimization problems. For multi-objective optimization problems, a tradeoff generally exists, which signifies that the improvement of one of the objectives comes at the expense of another. Multi-objective optimization problems do not comprise a single optimal design and comprise a set of designs representing the optimal tradeoffs among the goals. These tradeoffs are called Pareto optimal and can be visualized as a Pareto front. In this study, the Pareto optimal front is constructed by linking the trained surrogate models with a multi-objective optimization algorithm known as NSGA II.

An overview of the surrogates i.e., MLP and RSA is presented in the [Supplementary Materials](#). In addition, the theory and corresponding explanations regarding the construction and working of the single and multi-objective stochastic optimizations, i.e., PSO

Table 6
Correlations of the microscale CL model.

Description	Expression
Carbon volume fraction (ε_{Pt})	$\varepsilon_{Pt} = \frac{1}{\rho_{Pt}} \frac{L_{Pt}}{\delta_{CL}}$ (34)
Pt particle volume fraction (ε_C)	$\varepsilon_C = \frac{L_{Pt}}{\delta_{CL}} \left(\frac{1}{\rho_C wt\%} \right)$ (35)
Ionomer volume fraction (ε_e)	$\varepsilon_e = \left(\frac{1}{\rho_e} \right) \rho_C \varepsilon_C$ (36)
CL thickness (δ_{CL})	$\delta_{CL} = \left[\frac{(t) \rho_e}{\rho_C} \left(\frac{1}{\rho_C wt\%} \right) + \left(\frac{1}{\rho_{Pt}} + \frac{1}{\rho_C wt\%} \right) \right] L_{Pt}$ (37)
Thickness of the ionomer film (δ_e)	$\delta_e = \left[\left(1 + \frac{\rho_e}{\rho_C} \right)^{\frac{1}{3}} - 1 \right] r_C$ (38)
Thickness of water film (δ_w)	$\delta_w = \left[\left(1 + \frac{\rho_e}{\rho_C} + \frac{\rho_w}{\rho_C} \right)^{\frac{1}{3}} - 1 \right] r_C - \delta_e$ (39)
Oxygen balance in a single Pt/C particle domain	$C_{O_2}^g - C_{O_2}^{Pt} = -\Omega_T \frac{j}{4F a_C} = \Omega_T \frac{j}{4F \delta_{CL} a_C} = R_T \frac{j}{4F}$ (40)
Volumetric surface area of ionomer (a_C)	$a_C = \frac{3 \rho_C}{r_C} (r_C + \delta_e + \delta_w)^2$ (41)
Volumetric surface area of Pt (a_{Pt})	$a_{Pt} = \frac{a_{ECSA} L_{Pt}}{\delta_{CL}} \left(a_{ECSA} = \frac{3}{r_{Pt} \rho_{Pt}} \right)$ (42)
Total oxygen transport resistance (Ω_T)	$\Omega_T = \Omega_{w,int} + \frac{\delta_w^{eff}}{D_{O_2,w}} + \Omega_{e,int} + \frac{\delta_e^{eff}}{D_{O_2,e}} + \Omega_{Pt,int}$ (44)
Total transport resistance	$R_T = \frac{\Omega_T}{\delta_{CL} a_C}$ (45a)
Interfacial resistance in ionomer	$\Omega_{e,int} = Z_e \frac{\delta_e}{D_{O_2,e}}$ (45b)
Interfacial resistance in water film	$\Omega_{w,int} = Z_w \frac{\delta_w}{D_{O_2,w}}$ (45c)
Interfacial resistance in platinum particle	$\Omega_{Pt,int} = Z_{Pt} \frac{(r_C + \delta_e)^2}{r_C^2} \frac{\rho_{Pt}}{\rho_C} \left(\frac{r_C}{r_{Pt}} \right)^3 \left(\frac{1 - wt\%}{wt\%} \right) \frac{\delta_e}{D_{O_2,e}}$ (45d)
Effective ionomer thickness	$\delta_e^{eff} = \frac{(r_C + \delta_e + \delta_w)^2}{n_{Pt} r_{Pt}^2} \delta_e$ (46)
Effective water thickness	$\delta_w^{eff} = \frac{(r_C + \delta_e + \delta_w)^2}{n_{Pt} r_{Pt}^2} \delta_w$ (47)
Number of Pt particles on a single carbon particle	$n_{Pt} = \frac{\rho_C}{\rho_{Pt}} \left(\frac{r_C}{r_{Pt}} \right)^3 \left(\frac{wt\%}{1 - wt\%} \right)$ (48)
Oxygen concentration on the Pt particle surfaces ($C_{O_2}^{Pt}$)	$f(C_{O_2}^{Pt}) = B(C_{O_2}^{Pt})^{\frac{7}{4}} + \frac{4F}{\Omega_T} \frac{3 \rho_C}{r_C^2} (r_C + \delta_e)^2 C_{O_2}^{Pt} - \frac{4F}{\Omega_T} \frac{3 \rho_C}{r_C^2} (r_C + \delta_e)^2 C_{O_2}^g = 0$ where $B = \frac{3 L_{Pt}}{r_{Pt} \rho_{Pt} \delta_{CL}} \frac{1}{0.1} \left(\frac{1}{C_{O_2}^{ref}} \right)^{3/4} \exp\left(-\frac{E_c}{R} \left(\frac{1}{T} - \frac{1}{353.15} \right)\right) \exp\left(-\frac{z_e F \eta_c}{R T}\right)$ (49)

and NSGA II are given in the [Supplementary Materials](#) for interested readers.

4. Formulation of the multi objective optimization model

The electrocatalyst is one of the most expensive components of a PEMFC stack. The high cost of the cathode electrocatalyst is mainly due to the high catalyst loadings required to overcome the poor ORR activity constraints. Several studies have been conducted to maximize the PEMFC performance using advanced catalyst materials and structures. However, for MEAs, a suitable catalyst still needs to be discovered, particularly at the low Pt catalytic loadings required to meet the DOE's 2025 target of 0.1 mgPt/cm². Even with the progress of electrocatalyst technologies, these targets may be hampered for various reasons. Optimization of the cathode CL composition and structure to balance the proton and electron conducting phases as well as the optimization of size and distribution of pores for efficient reactant/product transport could be a way to meet the DOE's goal. Optimization through experiments is lengthy and time consuming, which may require several years. According to previous studies, the optimization through surrogate-based models has been often used to overcome this hindrance [35,38].

4.1. Selection of design variables and formulation of objective functions

In this study, to achieve the maximum cell voltage (V_{cell}) at a minimal cost of PEMFC fabrication and production (C_{stack}), the V_{cell} per individual cells and C_{stack} are considered as the objectives. The four important cathode CL parameters, i.e., L_{Pt} , $wt_{I/C}$, ε_{cCL} , and $wt_{Pt/C}$, are considered as the design variables. The lower to upper

ranges of design variables considered are 0.05 to 0.7 mg/cm² for L_{Pt} , 0.2 to 1.2 for $wt_{I/C}$, 0.2 to 0.7 for ε_{cCL} , and 0.2 to 1.2 for $wt_{Pt/C}$. Subsequently, the multi-objective optimization problem can be formulated as follows:

$$\text{Maximize } V_{cell} = f(L_{Pt}, wt_{I/C}, \varepsilon_{cCL}, wt_{Pt/C}) \quad (53)$$

$$\text{Minimize } C_{stack} = f(V_{cell}, L_{Pt}) \quad (54)$$

$$\text{subjected to } \begin{cases} L_{Pt}, 0.05 \leq L_{Pt} \leq 0.7 \text{ mg/cm}^2 \\ wt_{I/C}, 0.2 \leq wt_{I/C} \leq 1.2 \\ wt_{Pt/C}, 0.2 \leq wt_{Pt/C} \leq 1.2 \\ \varepsilon_{cCL}, 0.2 \leq \varepsilon_{cCL} \leq 0.7 \end{cases} \quad (55)$$

The C_{stack} in Eq. (56) represents the simplified cost model proposed by the DOE, USA [37]. The C_{stack} is a function of the area of MEA (A_{MEA}), L_{Pt} , the cost of Pt (C_{Pt}), and the number of cells in the stack (n_{cell}). A detailed explanation with regard to the parameters affecting C_{stack} is presented in [Section 4.3](#).

4.2. Surrogate construction

Once the optimization problem is formulated as described in the previous section, the next step is constructing the surrogate models as described in [Section 3](#). First, a set of sample points is drawn out using LHS technique. As shown in [Fig. S2a](#) and [b](#), the total number of sample points generated for this study is 85, and V_{cell} is numerically calculated at each sample point using the 3D two-phase multi-scale PEMFC model. The data obtained at each sample point together with V_{cell} are divided into training and test data; the data generated at 70 samples are used to design the surrogate model and 15 samples are used to test the surrogate model. Furthermore, in [Section 5.2](#), MLP and RSA are evaluated in terms of

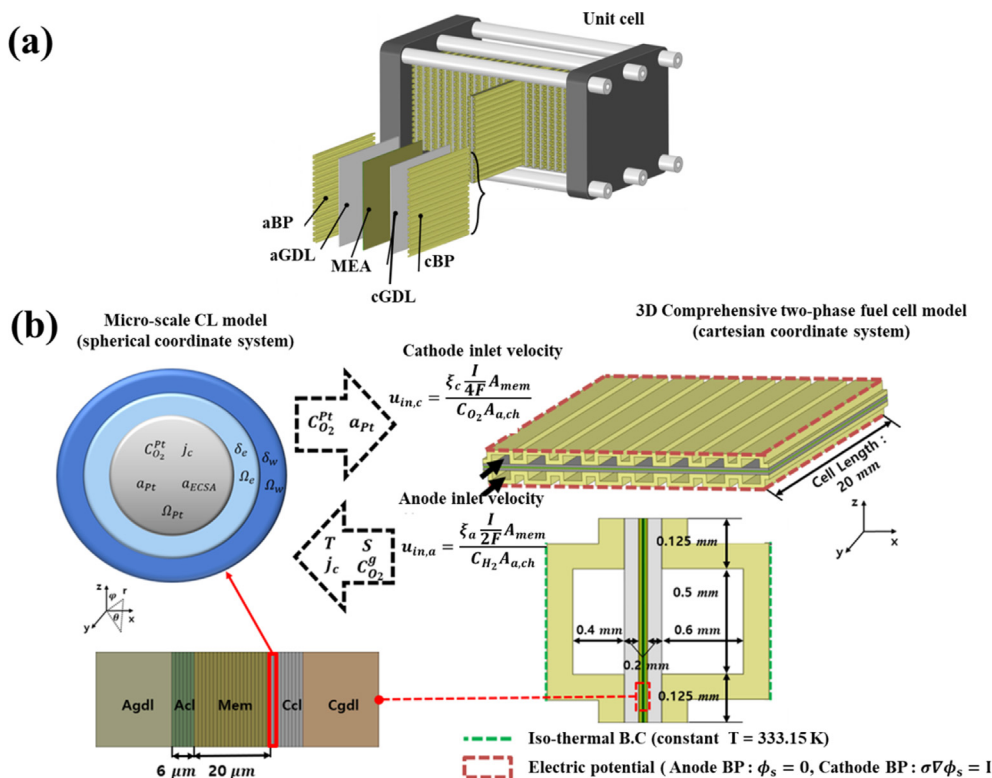


Fig. 1. (a) Schematic of the PEMFC stack and unit cell structures and (b) microscale and macroscale computational domains and switching variables for multi-scale PEMFC simulations.

their prediction accuracy using two metrics, adjusted R^2 and RMSE, and the obtained results are discussed.

4.3. Significance of reduction in catalyst costs over the PEMFC stack cost

As mentioned in the introduction, to make PEMFCs commercially viable, the manufacturing cost of the entire PEMFC system needs to be reduced. The two major components of an FC system are the FC stack and the balance of plant (BoP). BoP comprises the air loop, humidifier, water recovery loop, high and low temperature coolant loops, fuel loop, and FC system controller. To effectively evaluate the PEMFC system cost, a simplified cost model has been proposed by the DOE, USA [37], wherein the total system cost is divided into five subcategories and a scaling equation is provided for each subcategory. The five subcategories include PEMFC stack cost, C_{stack} , thermal management cost, humidification management cost, air management cost, fuel management cost, and BoP cost. Among the various component costs, the PEMFC stack cost contributes to 45% of the total system cost for producing between 100,000 and 500,000 systems/year.

As is well known, the changes in the production volumes significantly affect the manufacturing costs of PEMFCs. MEA is a major component of a PEMFC stack. It comprises of a PEM, two CLs, and two GDLs. Notably, when considering the MEA cost at low production volumes of around 1000 systems/year, the membrane cost dominates and attributes to 32% of the total cost, and the catalyst attributes to 16% of the total cost [49]. For production volumes of around 500,000 systems/year, the membrane cost is 11% whereas the catalyst cost drastically increases to 49% [49].

The aforementioned cost analysis of the total FC system provides insight into the need for reducing the cost of the catalyst

used in PEMFCs, especially for high production volumes (>500,000 units per year). This can be achieved by reducing the amount of catalyst loading per each CL. The DOE's simplified cost model considers a scaling equation for each individual cost component. The scaling equations are based on key system parameters of an individual component and are well known to the experts conducting the study. Considering this in mind and the effects of catalyst loading on the FC system cost, herein, we employ the simplified scaling equation proposed by the DOE for a typical 300 V PEMFC stack used in an automotive road vehicle. The simplified scaling equation to determine the total FC C_{stack} is as follows:

$$C_{stack} = 1.97 \times 10^{-4} \times ((0.16485 \times A_{MEA} \cdot n_{cell} + 588.33) \times L_{Pt} \times C_{Pt}) + (0.00883 \times A_{MEA} \cdot n_{cell}) + 342.03 \quad (56)$$

where the units of C_{stack} , A_{MEA} , and L_{Pt} are \$, cm^2 , and mg_{Pt}/cm^2 , respectively. The C_{Pt} value [mg_{Pt}] is determined based on the average cost of Pt over the last six months, i.e., 1021 /troy ounce (average C_{Pt} during January–July 2022). Note that Eq. (56) is constructed based on regression analysis and hence can be applied only for a given range of input parameters. The range of input parameters described by the DOE is illustrated in Table 7.

5. Results and discussion

In this study, the results of the optimization framework are presented. In Section 5.1, the results of the reference case validated against the experimental results are described. In Section 5.2, the prediction performance of the surrogate models, namely MLP and RSA, is discussed. In Section 5.3, a comprehensive analysis of the key micro-scale cathode CL parameters affecting V_{cell} is presented.

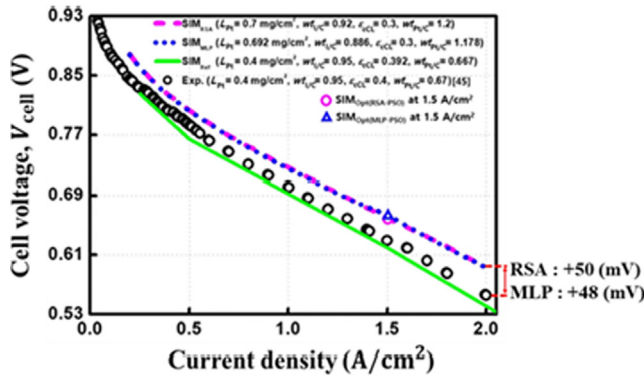


Fig. 2. Comparison of the simulated and measured polarization curves under anode and cathode conditions of 2/2 bar, 1.2/2.0 stoichiometry, and 100%/100% inlet relative humidity. The cell temperature is 353.15 K. The experimental data were given by Liu et al. [50]. The cathode CL design for both Exp. and Sim_{Ref} is based on $L_{Pt} = 0.4 \text{ mg/cm}^2$, $wt_{Pt/C} = 0.95$, $wt_{Pt/C} = 0.67$, and $\varepsilon_{CL} = 0.4$.

Table 7

Range of input parameters to analyze the validity of the PEMFC stack cost model using Eq. (56) [37].

Input parameters	Range	
	Lower bound	Upper bound
Stack power, [kW]	60	120
Stack voltage, V_{stack} [V]	100	300
Pt loading, L_{Pt} [mg/cm ²]	0.1	0.8
Total MEA area, $A_{MEA} \cdot n_{cell}$ [cm ²]	56,080	112,147
Pt cost, C_{Pt} [mg]	800	2,000

Furthermore, in Section 5.4, detailed insight into the effects of the key micro-scale parameters on V_{cell} and C_{stack} considering a multi-objective optimization problem are described.

5.1. Experimental validation of the PEMFC model

To obtain reliable PEMFC data for the multi-objective optimization study, the 3D PEMFC model described in Section 2 is first validated using the experimental data given by Liu et al. [50], who considered $L_{Pt} = 0.4 \text{ mg/cm}^2$, $wt_{Pt/C} = 0.95$, $\varepsilon_{CL} = 0.4$, and $wt_{Pt/C} = 0.67$ for their CL fabrication, which are well within the ranges of the CL design variables in Eq. (54). As shown in Fig. 2,

the PEMFC model accurately captures the polarization characteristics of the experimental cell, clearly demonstrating the accuracy and validity of the present model.

5.2. Evaluation of surrogate prediction performance

Once the optimization problem is formulated and the surrogate models are constructed as described in Sections 4.1 and 4.2, they are further evaluated in terms of their prediction capabilities. To provide further insight into the constructed surrogates, the corresponding weights for MLP and the second-order equation of RSA are presented in Table 8. In this study, MLP is constructed by considering a single hidden layer comprising 10 nodes.

The initialization of the weight in the layers is performed considering the RMSprop and Xaviers initialization algorithms. Furthermore, the prediction capabilities of the trained surrogate models are evaluated through an error analysis, as described in Section 3.3, and the error analysis results are plotted in Fig. 3. For the trained MLP and RSA, the adjusted R^2 values obtained are 0.9890 and 0.9808 and the RMSE values are 5.1 and 7.6 mV, respectively. Since the adjusted R^2 values are close to 1 and RMSE values are low enough to indicate a good fit, we can conclude that both MLP and RSA are capable of adapting to highly nonlinear data. However, to determine the capabilities of the trained surrogate models, their prediction capability is evaluated at a set of unknown points. Hence, a set of 15 randomly generated test points is established via the LHS technique, and the surrogate models are tested at these points. The error plots of the tested surrogate models are plotted in Fig. 3(c), wherein the obtained adjusted R^2 and RMSE values show that MLP surrogate exhibit better and more reliable prediction abilities than RSA. In terms of the adjusted R^2 values, MLP performs better than RSA by 0.0385, while the RMSE value of MLP is smaller than that of RSA by 6.7 mV. The difference in the prediction ability between the two surrogate models can be attributed to the fact that MLPs are constructed using neural networks and involve a complex learning process of feed-forward and back propagation, whereas RSAs are simple surrogate models that are based on curve fitting and consider a second-order quadratic equation for the fitting process. Hence, MLP provides more reliable results for a given set of highly nonlinear data than RSA.

5.3. Single-objective optimization for PEMFC performance

Subsequently, once the RSA and MLP models are tested and are deemed fit for further analysis, they are integrated with the PSO

Table 8

Nodal weights constituting MLP and the corresponding second order equation for the RSA surrogate.

Node number	Input - Hidden				Hidden - Output	
	L_{Pt}	$wt_{Pt/C}$	$wt_{Pt/C}$	ε_{CL}	Node number	Output
1	-1.452	-15.219	-1.049	7.259	1	-0.595
2	-0.509	3.068	0.316	-3.243	2	1.562
3	-3.702	-4.894	1.174	0.597	3	0.919
4	-1.302	1.427	0.070	0.048	4	-0.802
5	-2.055	5.273	-0.82	-6.596	5	-0.373
6	-0.140	-3.899	1.485	-0.070	6	1.449
7	3.693	-2.064	-8.012	-0.558	7	-0.223
8	-16.142	1.591	0.484	-1.995	8	-0.452
9	-1.443	-3.039	2.812	-3.15	9	-2.202
10	-3.266	-1.929	5.499	-7.291	10	0.503

RSA

$$y = 0.4795 + 0.1463x_1 + 0.1748x_2 + 0.0794x_3 + 0.0277x_4 - 0.1002x_1^2 - 0.1613x_2^2 - 0.0546x_3^2 - 0.1918x_4^2 + 0.0546x_1 \cdot x_2 + 0.0366x_1 \cdot x_3 - 0.0864x_1 \cdot x_4 - 0.0184x_2 \cdot x_3 + 0.1038x_2 \cdot x_4 + 0.0591x_3 \cdot x_4$$

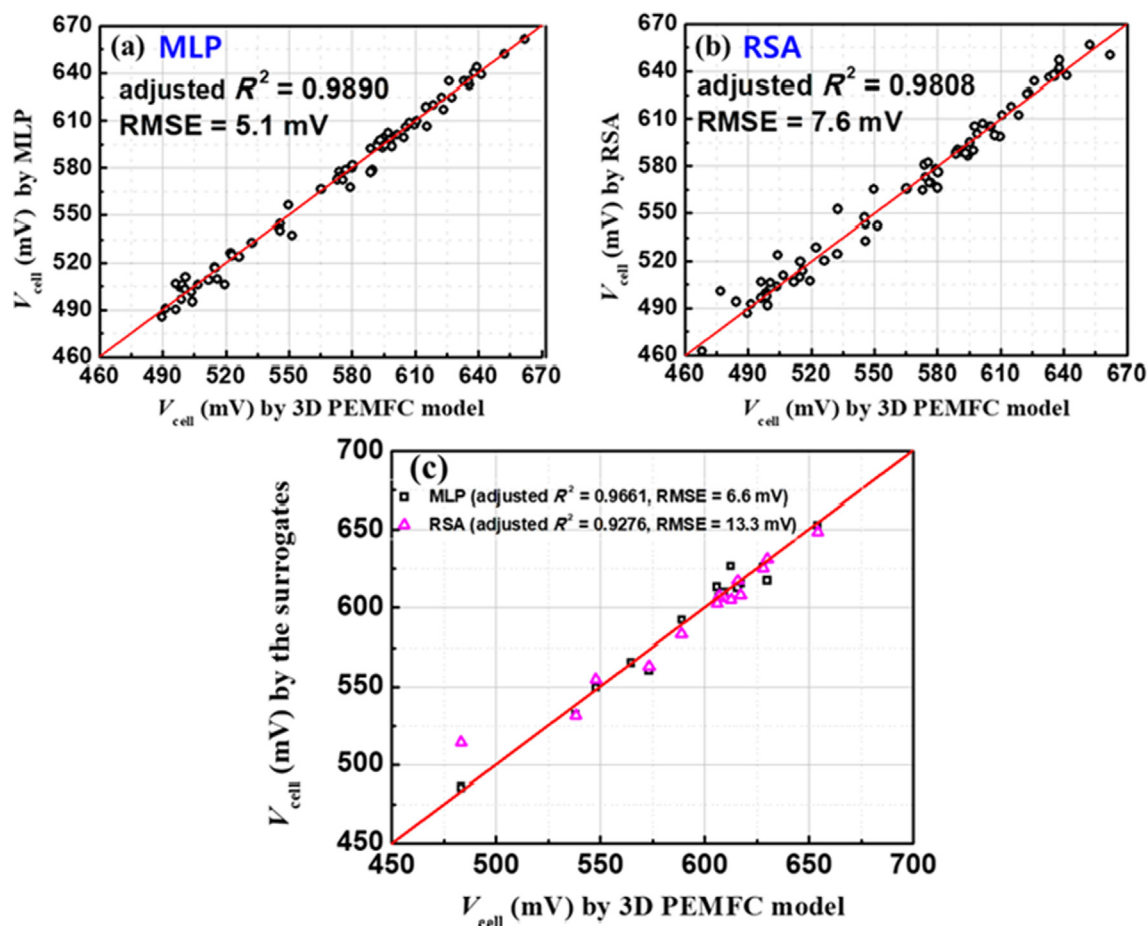


Fig. 3. Scatter plots for the training data (70 Points) for (a) MLP and (b) RSA, and (c) the scatter plots for the test data (15 points).

algorithm for single-objective optimization, wherein the optimal values of the CL design variables are searched to maximize the cell voltage (V_{cell}) only without considering the C_{stack} . As PSO is based on an iterative updating strategy with the particle position, and during calculations, both the random initial population and the maximum number of iterations are set to 1000 at the beginning of the optimization algorithms. Fig. 2 compares the V_{cell} based on the reference CL design and the optimum CL designs obtained from RSA and MPL (symbols at $I = 1.5 \text{ A/cm}^2$). In addition to the surrogate model predictions, the PEMFC simulations using the optimum design variables of RSA and MLP are performed in a wide range of current density up to 2.0 A/cm^2 for comparison purpose. First, small differences between the PEMFC simulation and individual surrogate models are observed, i.e., around 2.9 mV for RSA and 7.0 mV for MLP at $I = 1.5 \text{ A/cm}^2$. More importantly, as compared with the reference CL design, the superior V_{cell} was predicted by the PEMFC simulations to which the optimum CL designs found from RSA and MLP were applied. For instance, the V_{cell} improvement at 2.0 A/cm^2 is around 50 and 48 mV using the optimum CL designs of RSA and MLP, respectively.

5.4. Effects of cathode CL design variables on the cell performance, V_{cell}

To comprehend the precise impact of each design variable on the cell performance, the model predictions made using the MLP approach are provided in Fig. 4. These graphs are plotted for each design variable vs. V_{cell} by setting the three design variables to their optimum levels. For instance, in Fig. 4(a), to examine the impact of altering L_{Pt} on V_{cell} , the other design variables are set

to their optimal values ($wt_{I/C} = 0.886$, $\varepsilon_{cCL} = 0.3$, and $wt_{Pt/C} = 1.178$). As shown in Fig. 4(a), V_{cell} increases with L_{Pt} . This is because as L_{Pt} increases, the electrochemically active area of three phase boundary (Pt/ionomer/air) increases to a certain extent and promotes the ORR rate. However, V_{cell} is on the lower side for a lower L_{Pt} as the total number of reaction sites for ORR is reduced. Additionally, Fig. 4(a) shows that for $L_{Pt} < 0.2 \text{ mg/cm}^2$, the slope of V_{cell} with respect to L_{Pt} , is steeper indicating that the loss in catalyst utilization is more significant in the low Pt loading regime. In contrast, the improvement in V_{cell} with increasing L_{Pt} is less significant for $L_{Pt} > 0.2 \text{ mg/cm}^2$, signifying that the addition of Pt has a relatively minor impact on V_{cell} . Therefore, clearly, the reduction of L_{Pt} to 0.2 mg/cm^2 is effective under the given CL design condition ($wt_{I/C} = 0.886$, $\varepsilon_{cCL} = 0.3$, and $wt_{Pt/C} = 1.178$), and increasing L_{Pt} on a large scale is not technically justified.

Regarding the influence of $wt_{I/C}$ in Fig. 4(b), there is a tradeoff with the $wt_{I/C}$ being used in the cathode CL. When the $wt_{I/C}$ is below the optimum value ($wt_{I/C} = 0.886$), low ionomer values cause a decrease in the proton conduction and hence a loss in V_{cell} . For $wt_{I/C} > 0.886$, an increase in $wt_{I/C}$ would be ideal for efficient proton transport; however, excess ionomer would lead would cause the thickness of the ionomer film to increase and thus hinder the penetration of oxygen toward the Pt surface, resulting in a drop in V_{cell} . With regard to the effects of ε_{cCL} on V_{cell} in Fig. 4(c), the optimum value of ε_{cCL} is around 0.3. Note that under the condition of constant L_{Pt} , $wt_{I/C}$, and $wt_{Pt/C}$, the cathode CL thickness increases from 7.593 to $20.248 \text{ }\mu\text{m}$ with increasing ε_{cCL} . As ε_{cCL} increases beyond 0.35, V_{cell} significantly decreases, which is mainly due to the thick cathode CL and causes a more profound effect of Ohmic

loss. V_{cell} slightly decreases when ε_{cCL} decreases below 0.3; this is indicative of increased oxygen transport resistance with decreasing ε_{cCL} . In terms of the influence of $wt_{\text{Pt/C}}$ on V_{cell} when the other variables are constant ($L_{\text{Pt}} = 0.692 \text{ mg/cm}^2$, $wt_{\text{I/C}} = 0.886$, and $\varepsilon_{\text{cCL}} = 0.3$; Fig. 4d), a lower $wt_{\text{Pt/C}}$ carries a significantly larger amount of carbon powder and results in a thicker cathode CL, which consequently hinders the diffusion of oxygen and thereby lowers catalyst utilization and cell performance. In contrast, the larger $wt_{\text{Pt/C}}$ yields a thinner CL, introducing more Pt particles per unit CL volume and leading to improved cell performance. The cathode CL thickness varies from 48.85 to 8.527 μm under the limited constraint condition of $wt_{\text{Pt/C}}$ ranging between 0.2 and 1.2, wherein V_{cell} relatively slightly varies around 67 mV compared to the other design variables shown in Fig. 4(a–c).

The aforementioned discussions clearly denote that L_{Pt} has a more significant effect on V_{cell} than the other variables. To better understand the effects of interaction between L_{Pt} and the design variables of $wt_{\text{I/C}}$ and ε_{cCL} , contour plots are presented in Fig. 5. In Fig. 5(a), the impact of altering L_{Pt} and $wt_{\text{I/C}}$ on V_{cell} is shown by keeping the other two design variables ($wt_{\text{Pt/C}} = 1.178$ and $\varepsilon_{\text{cCL}} = 0.3$) constant. For lower L_{Pt} , Fig. 5(a) shows a noticeable decrease in V_{cell} at higher $wt_{\text{I/C}}$ values, clearly indicating the considerable effect of oxygen transport limitation through the ionomer film in the cathode CL, i.e., the film thickness increases with the $wt_{\text{I/C}}$ value. In contrast, the maximum V_{cell} predicted by MLP is 671 mV when both L_{Pt} and $wt_{\text{I/C}}$ are high. Under the high Pt loading condition, where more Pt nanoparticles are distributed on each carbon particle, the actual oxygen transport path across the ionomer film is short, and hence, the effect of the ionomer film thickness on V_{cell} is less

significant, which clearly indicates the dominant effect of proton transfer over oxygen transfer through the ionomer film under high Pt loading conditions. Fig. 5(b) demonstrates the interactions of L_{Pt} and ε_{cCL} and their impact on V_{cell} . A higher V_{cell} value is obtained when ε_{cCL} is between 0.2 and 0.4. Interestingly, the optimal range of ε_{cCL} shifts toward a higher porosity regime when L_{Pt} decreases. Since the CL design with higher ε_{cCL} values lead to the lower content of ionomer and hence thinner ionomer film thickness, the CL performance under the low L_{Pt} condition is mainly determined by the oxygen transfer through the ionomer film, while the high L_{Pt} design diminishes the oxygen transfer effect.

5.5. Multi-objective optimization for PEMFC performance and cost

The CL design variables considered in this study significantly impact V_{cell} . Although the higher L_{Pt} results in the higher V_{cell} , it increases C_{stack} . Consequently, a tradeoff occurs between V_{cell} and C_{stack} , which makes a multi-objective optimization necessary. Hence, in the further discussion presented, the impact of the design variables, i.e., L_{Pt} , $wt_{\text{I/C}}$, $wt_{\text{Pt/C}}$, and ε_{cCL} , on both V_{cell} and C_{stack} is analyzed. Accordingly, the multi-objective optimization problem is formulated as described in Section 4.1. In the multi-objective problem, V_{cell} must be maximized and C_{stack} must be simultaneously minimized. The objective functions in such a problem often conflict with each other. From Eqs. (53), (54), and (56), C_{stack} increases with V_{cell} . As the solution to the multi-objective problem is a set of points representing the best tradeoffs between the objective functions for each solution, an objective function cannot be further improved without making at least one other objec-

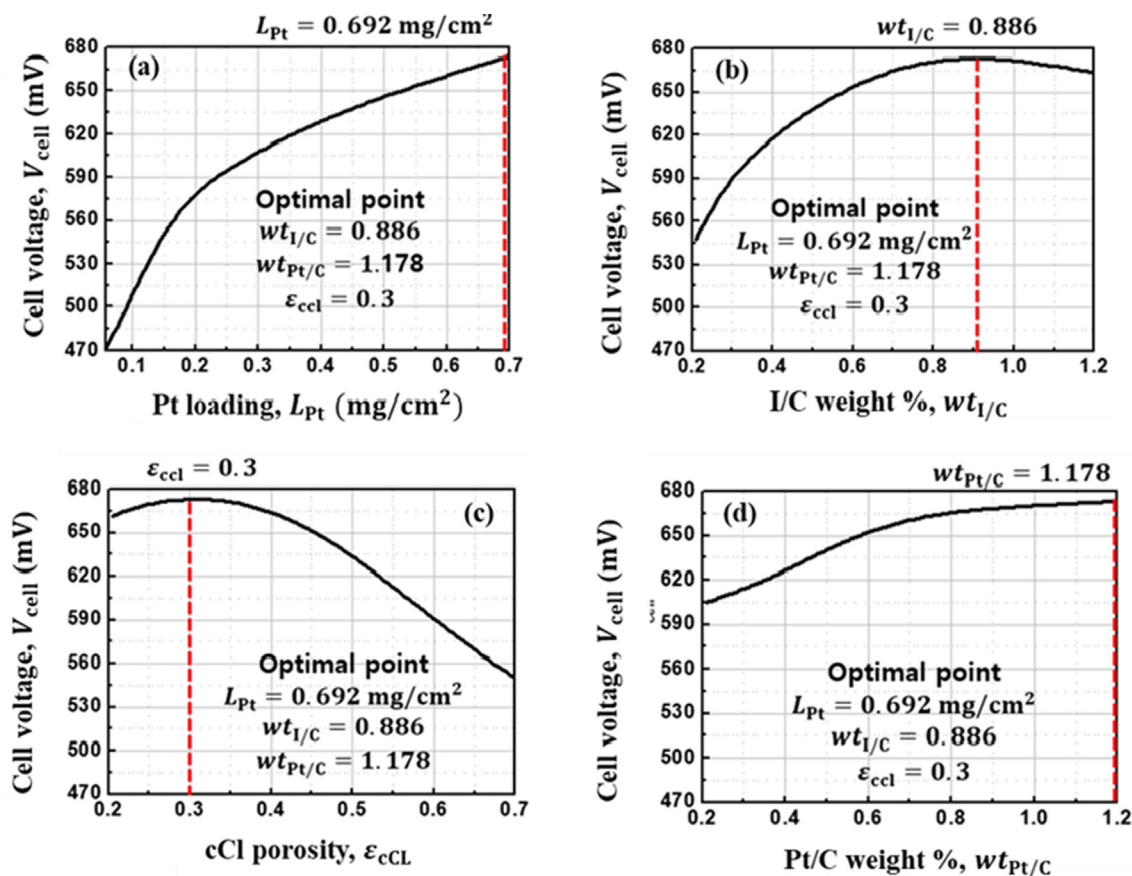


Fig. 4. Effects of cathode CL design variables on the cell performance, V_{cell} based on MLP prediction at $I = 1.5 \text{ A/cm}^2$.

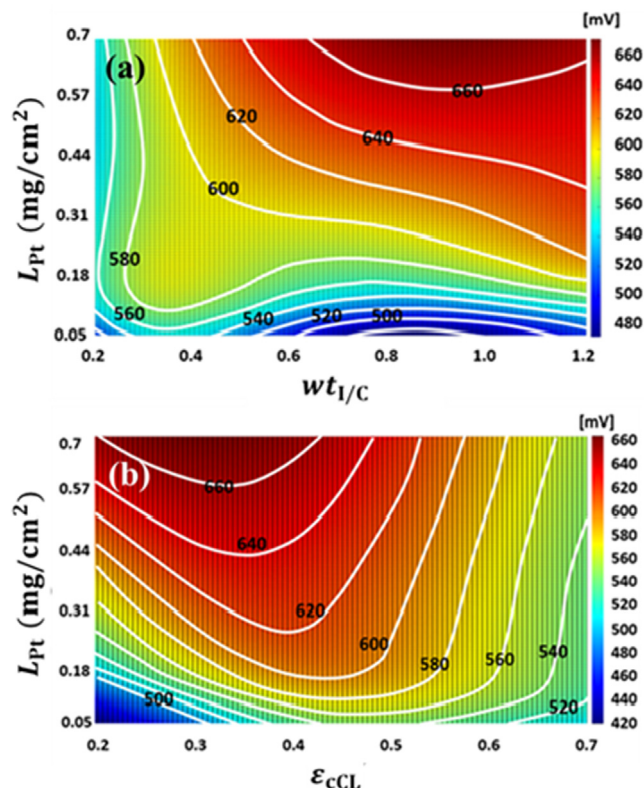


Fig. 5. Performance maps with regard to two CL design variables based on the MLP surrogate predictions at $I = 1.5 \text{ A/cm}^2$, wherein the other two CL design variables are fixed at (a) $\varepsilon_{\text{cCL}} = 0.3$, $wt_{\text{Pt/C}} = 1.178$ and (b) $wt_{\text{Pt/C}} = 1.178$, $wt_{l/c} = 0.886$.

tive function worse. These points are known as Pareto optimal points. The Pareto optimal set or Pareto front is the collection of all Pareto optimal points. Herein, NSGA II was constructed in-house using MATLAB R2021a and then linked to MLP to solve the multi-objective optimization algorithm.

To begin the optimization algorithm, first, a random initial population is generated that is within the bounds of the design variables. With the initial conditions of the design variables set, the corresponding trade-offs or Pareto solutions must be determined. Since predicting how many iterations are necessary to achieve a desirable Pareto front is challenging for decision makers, caution is frequently adopted. Moreover, the number of required iterations must be kept to a minimum to prevent needless calculation time loss. In Fig. 6, the process of Pareto optimal front formation for the multi-objective formulation problem considering a range of iterations between 30 and 750 is realized. Fig. 6 shows that the solutions generated at the end of 30 iterations are randomly distributed within the feasible space. Furthermore, as the number of iterations increases from 50 to 250, the generated solutions advance closer to the Pareto front via successive selection, mutation, and crossover processes during the following generations. At the end of 500 iterations, the solutions converge, and all the updated points are dominant and exhibit the best diversity, i.e., the points are spread along both the objective functions and evenly distributed. Hence, a further increase in the number of iterations (750) will not cause any further improvement in the desired solution. Therefore, 500 iterations are deemed as the optimal stopping criteria to assess the performance of the optimization algorithm and save computational time.

In Fig. 7(a), the Pareto optimal fronts for V_{cell} and C_{stack} obtained by the multi-objective optimization are compared with the sample data, the reference point, and the single-objective optimal point

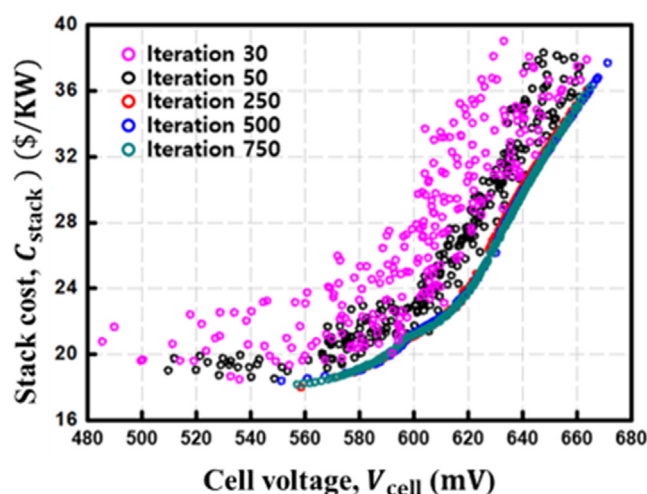


Fig. 6. Scatter plot of stack cost (C_{stack}) versus cell voltage (V_{cell}) under different iteration numbers to obtain the pareto sets. All calculations for V_{cell} and C_{stack} are based on $I = 1.5 \text{ A/cm}^2$.

($Opt_{\text{MLP-PSO}}$), i.e., calculated by combined MLP and PSO algorithms. The figure shows that the solutions of the Pareto front exhibit a clear tradeoff between V_{cell} and C_{stack} and lie below the sample data and reference points, clearly indicating that the optimal CL design points result in superior cell performance at a relatively lower unit price. Additionally, the single-objective optimal point is located at the point where both V_{cell} and C_{stack} are the highest on the Pareto optimal front. The optimum values of CL design variables constituting the Pareto front are normalized as follows and plotted in Fig. 7(b).

$$x_{\text{normalized}} = \frac{x_{\text{opt}} - x_{\text{low}}}{x_{\text{high}} - x_{\text{low}}} \quad (57)$$

where x_{low} and x_{high} represent the lower and upper limits for individual CL variables given in Eq. (56), respectively. The curve of L_{Pt} is similar to the Pareto optimal front curve in Fig. 7(a), signifying that L_{Pt} is a decisive factor that heavily influences both V_{cell} and C_{stack} . In contrast, the CL porosity ε_{cCL} has a relatively small impact on V_{cell} and C_{stack} , wherein a mild decrease in ε_{cCL} is required as V_{cell} and C_{stack} increase. Notably, two optimum ranges exist for both $wt_{l/c}$ and $wt_{\text{Pt/C}}$. For the lower V_{cell} and C_{stack} regime, lower optimum values of $wt_{l/c}$ and $wt_{\text{Pt/C}}$ are required because the oxygen permeation through the ionomer film is a dominant factor under lower Pt loading conditions and hence less number of Pt particles on each carbon particle in the cathode CL. On the other hand, in the higher V_{cell} and C_{stack} regime, representing higher L_{Pt} , proton transfer through the ionomer is more critical than oxygen permeation and the optimal design point for $wt_{l/c}$ and $wt_{\text{Pt/C}}$ is formed at their higher values.

To validate the results obtained by the multi-optimization, two optimal CL design points designated as Opt. 1 and Opt. 2 are selected and simulated by the 3D PEMFC model. As indicated in Fig. 7(a), Opt. 1 has the same V_{cell} of 620 mV with the reference point, while Opt. 2 is chosen based on the similar stack price of around 29/kW. Table 9 compares the simulation results of V_{cell} and C_{stack} using the optimum values of the four CL design variables. Compared to the reference CL design, the Opt. 1 design provides a cost reduction of around \$5.86/kW, while maintaining the same cell performance of $V_{\text{cell}} = 620 \text{ mV}$, which is due mainly to the reduction in the amount of Pt used ($L_{\text{Pt}} = 0.242 \text{ mg/cm}^2$) compared to the reference design ($L_{\text{Pt}} = 0.4 \text{ mg/cm}^2$). When C_{stack} remains the same around 29/kW, the improvement in V_{cell} via using the CL

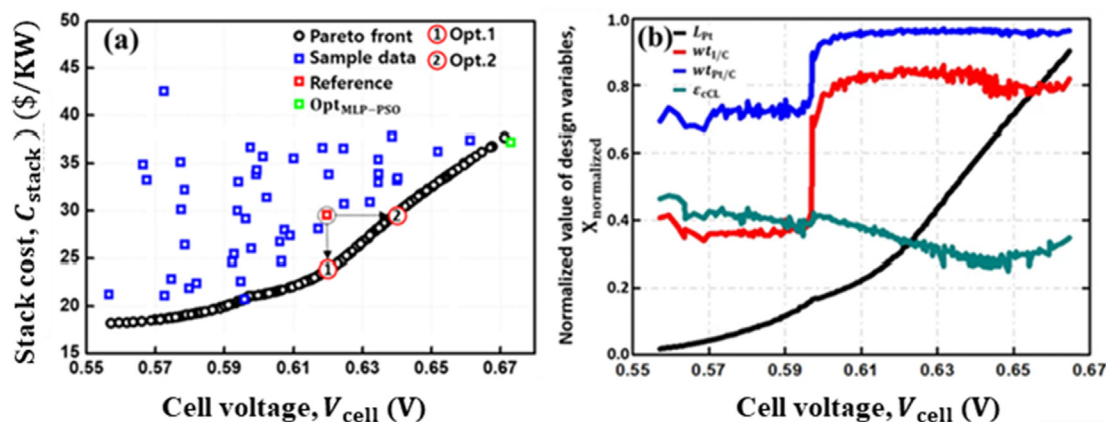


Fig. 7. (a) Comparison of the Pareto optimal front of two objectives (C_{stack} and V_{cell}) with the sample data, reference point, and single-objective optimal point by PSO. The Pareto front curve comprises individual 250 optimal points corresponding to different CL designs and (b) normalized CL design variables constituting the Pareto-optimum front. All calculations for V_{cell} and C_{stack} are based on $I = 1.5 \text{ A/cm}^2$.

Table 9

Comparison of optimal CL designs via single- and multi-objective optimization with the reference CL design. All calculations for V_{cell} and C_{stack} are based on $I = 1.5 \text{ A/cm}^2$.

Opt.	L_{Pt} [mg/cm ²]	$wt_{I/C}$	ϵ_{cCL}	$wt_{Pt/C}$	V_{cell} [mV]	C_{stack} [/kW]
Opt. 1 (Multi-objective)	0.242	1.040	0.373	1.165	618	23.68
Opt. 2 (Multi-objective)	0.421	1.022	0.333	1.167	648	29.24
Reference design	0.4	0.95	0.392	0.6667	620.8	29.536
Opt _{MLP-PSO} (Single-objective)	0.692	0.886	0.3	1.178	664	38.184

design of Opt.2 is 28 mV. Note that the L_{Pt} for Opt. 2 ($L_{Pt} = 0.421 \text{ mg/cm}^2$) is higher than that of the reference design ($L_{Pt} = 0.4 \text{ mg/cm}^2$), but C_{stack} is approximately the same. According to Eq. (56), C_{stack} is directly related to the total number of cells in the stack (n_{cell}) and L_{Pt} . Owing to the higher V_{cell} in Opt. 2 compared to that in Opt. 1 at the same operating current density ($I = 1.5 \text{ A/cm}^2$), the n_{cell} required to construct a PEMFC stack with the same power is reduced. Therefore, a tradeoff occurs between L_{Pt} and n_{cell} , which makes the C_{stack} for the reference design and Opt. 2 identical.

6. Conclusions

This study employed two famous surrogate models, namely, MLP and RSA, with optimization algorithms PSO and NSGAII to determine the optimal cathode CL parameters: L_{Pt} , $wt_{I/C}$, $wt_{Pt/C}$, and ϵ_{cCL} . First, the surrogate models were constructed and tested based on the simulation results of a 3D, multi-scale, two-phase PEMFC model. Then, the prediction capabilities of the surrogate models were compared considering two measures of performance: RMSE and adjusted R^2 . According to the performance measures obtained using 15 randomly generated test samples, MLP outperformed RSA and the accuracy of RSA was relatively poor. Furthermore, the MLP and RSA surrogate models were integrated with the PSO algorithm for single-objective optimization, wherein the optimal values of the cathode CL parameters were successfully obtained to maximize the cell performance (V_{cell}). Compared with the reference CL design, superior V_{cell} was predicted by the PEMFC simulations to which the optimum CL designs found from MLP and RSA were applied. At $I = 1.5 \text{ A/cm}^2$, the V_{cell} improvement was around 50 and 48 mV using the optimum CL design of MLP and RSA, respectively.

The improvement in V_{cell} as observed from the single-objective optimization was mainly due to higher L_{Pt} leading to an increase in

the overall PEMFC stack price (C_{stack}). Therefore, multi-objective optimization was performed with MLP integrated with the NSGAII optimization algorithm. In the multi-objective optimization, the objective function of V_{cell} was maximized while that of C_{stack} was minimized. The Pareto optimal front for V_{cell} and C_{stack} obtained by the multi-objective optimization indicated that a clear tradeoff exists between V_{cell} and C_{stack} . Compared to the reference CL design, a cost reduction of around \$5.86/kW was achieved at $V_{cell} = 620 \text{ mV}$ and $I = 1.5 \text{ A/cm}^2$, which is mainly due to the reduction in the amount of Pt used ($L_{Pt} = 0.242 \text{ mg/cm}^2$) compared to that in the reference design ($L_{Pt} = 0.4 \text{ mg/cm}^2$). When C_{stack} was fixed as 29/kW, the improvement in V_{cell} was around 28 mV at $I = 1.5 \text{ A/cm}^2$. This study clearly demonstrates the capability of using surrogate models and optimization algorithms to mitigate the hindrance of expensive and time-consuming optimization procedures of cathode CL parameters in PEMFCs.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary material

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jechem.2023.02.027>.

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